ACTL2102 Tutorial

Week 5: The Poisson Process

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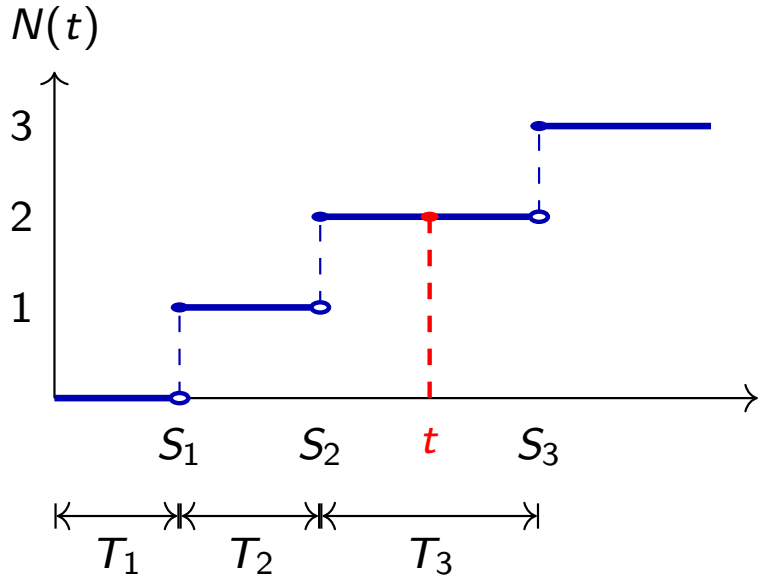
T3 2026

Plan

- Recap: interarrival and waiting times, the conditional distribution of arrival times, the non-homogeneous Poisson process, superposition and thinning
- Main questions: **3.5, 3.6, 3.9, 3.11, 3.15**
- If time permits: 3.13, 3.14

Interarrival and Waiting Times

Let T_n be the time between the $(n-1)$th and nth events of a Poisson process with rate λ, and S_n the time of the nth event.



The **interarrival times** $T_1, T_2, \ldots$ are i.i.d. Exp(λ), each with mean $1/\lambda$.

The **waiting time** $S_n = T_1 + \cdots + T_n$ is Gamma(n, λ), with mean n/λ and variance n/λ^2.

The nth event occurs after time t if and only if fewer than n events occur by time t. In the figure, $N(t) = 2$ and $S_3 > t$.

$$\{S_n > t\} \iff \{N(t) < n\}, \quad \text{so} \quad \Pr(S_n > t) = \sum_{k=0}^{n-1} e^{-\lambda t} \frac{(\lambda t)^k}{k!}.$$

With $n = 1$, $\Pr(T_1 > t) = \Pr(N(t) = 0) = e^{-\lambda t}$, so $T_1 \sim \text{Exp}(\lambda)$.

Exercise 3.11: Interarrival times: at least two customers in ten minutes

Exercise 3.11: [poi7, Solution] The time elapsed between arrivals of customers is modeled such that V_k represents the time elapsed between the arrivals of the $k-1^{\text{th}}$ and k^{th} customer. (V_1 = the time the first customer arrives). Suppose that:

i) $V_1, V_2, \ldots$ Are mutually independent;

ii) The p.d.f. of each V_k is $f(t) = 0.1e^{-0.1t}$, for $t > 0$, where t is measured in minutes.

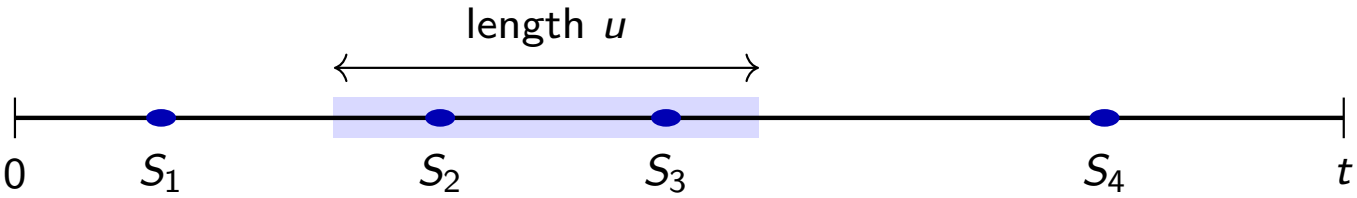
Determine the probability that at least two customers arrive in a ten-minute period.

Conditional Distribution of Arrival Times

Theorem

Given $N(t) = n$, the arrival times $S_1, \ldots, S_n$ are distributed as the order statistics of n independent $U(0, t)$ random variables:

$$f(s_1, \ldots, s_n \mid N(t) = n) = \frac{n!}{t^n}, \qquad 0 < s_1 < \cdots < s_n < t.$$



Given $N(t) = n$, the events are located as n independent $U(0, t)$ points, and S_k is the kth smallest of these points. Each point lies in a given sub-interval of length u with probability u/t, so the number of events in the sub-interval is $\text{Bin}(n, u/t)$.

Example. Given $N(1) = 3$, the number of events in $(0, \frac{1}{4}]$ is $\text{Bin}(3, \frac{1}{4})$, so all three occur there with probability $(\frac{1}{4})^3 = \frac{1}{64}$.

Exercise 3.6: Burglaries: first and second waiting times

Exercise 3.6: [poi2, Solution] The number of burglaries in a Poisson process with expected number of burglaries 2 per year.

1. What is the probability that the time to the first burglary is more than 1 year?
2. What is the probability that waiting time till the second burglary will be longer than a year?

The expected number of burglaries was estimated using ten years of data. Two years ago the street decided to have a Residential Security System.

3. What is the probability that there will be 20 burglaries in the first eight years and then no burglaries in two succeeding years, given that the total number of burglaries is 20. Do you need to assume that the rate of burglaries is estimated correctly?
4. The provider of the Residential Security System advertises with this example that the system works. Do you find sufficient evidence to support this claim?

Exercise 3.5: Accident proneness: conditioning on an unknown rate

Exercise 3.5: [poi1, Solution] Accident-proneness may be assumed to be measured by a parameter $b \in \{1, 2, 3, \ldots\}$. An individual with accident-proneness b will suffer accidents according to a Poisson process with rate λb per year, where λ is known. The proportion of the population with accident-proneness b is $\pi(b)$ for each b.

1. Write down the probability of suffering x accidents in t years given b.
2. If a particular individual has experienced x accidents in t years, calculate the probability that that individual has accidents-proneness b. Hence find an expression for the probability that the specified individual will not have an accident between time t and time $t + 1$.

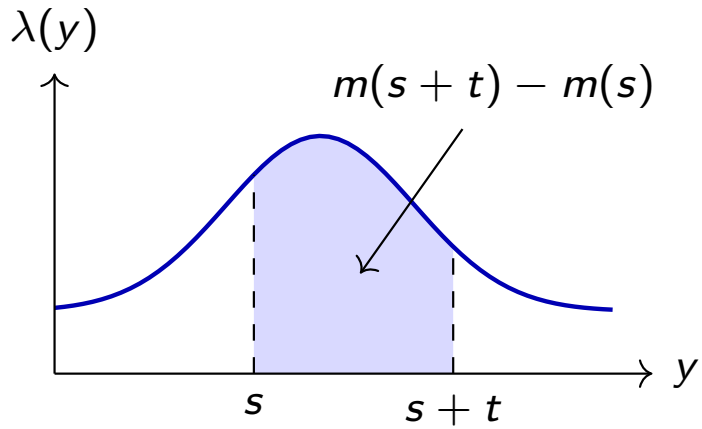
The Non-Homogeneous Poisson Process

A counting process $\{N(t), t \geq 0\}$ is a **non-homogeneous Poisson process** with **intensity function** $\lambda(t)$, $t \geq 0$, if $N(0) = 0$, it has independent increments, and as $h \to 0$

$$\Pr[N(t+h) - N(t) = 1] = \lambda(t)\, h + o(h),$$
$$\Pr[N(t+h) - N(t) \geq 2] = o(h),$$

where a function f is $o(h)$ if $\lim_{h \to 0} f(h)/h = 0$.
The **mean value function** is

$$m(t) = \int_0^t \lambda(y)\, dy.$$



$N(s+t) - N(s)$ is Poisson with mean $m(s+t) - m(s)$, the area under λ over $(s, s+t]$.
The increments are independent but in general not stationary: the distribution of $N(s+t) - N(s)$ depends on s as well as on t.

Exercise 3.9: Two lines of business: one homogeneous, one not

Exercise 3.9: [poi5, Solution] An insurer has two lines of business. In the first line of business claims arrive according to a homogeneous Poisson process with rate $\lambda_1 = \lambda$. In the second line of business claims arrive according to a nonhomogeneous Poisson process with time dependent rate $\lambda_2(t) = 2t+2$.

1. Show that mean value function for claims of the second line of business is given by:

$$m_2(t) = t^2 + 2t, \qquad \text{for } t \geq 0$$

2. Show that mean value function for claims of the first line (homogenous Poisson process) of business is given by:

$$m_1(t) = \lambda t, \qquad \text{for } t \geq 0$$

3. For each of the lines of business, what is the probability that n events occur between time $t = 4$ and $t = 5$?

4. For each of the lines of business, what is the probability that n events occur between time $t = 4$ and $t = 6$?

Exercise 3.15: Accident rate declining with driving experience

Exercise 3.15: [poi11, Solution] The number of car accidents an insured person faces depends on their experience. The rate at which Liam will have car accidents is given by:

$$R(e) = \frac{(e+a)^{-b}}{c},$$

where e is the experience (in years), $a = 0.5$ is experience before obtaining a drivers licence (time in driver school), $b = 2$ is the speed of learning, and $c = 200$ represents the quality of the driver.

1. Find the probability that Liam, who just obtained his drivers licence, will have at least one accident in the:
   (a) first year.
   (b) second year.
   (c) first 10 years.
2. Liam tells the insurance company that he obtained his drivers licence ten years ago and had only one accident. What is the probability that Liam had his claim after the first year of obtaining his drivers licence?

Superposition and Thinning

Superposition. If $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ are independent Poisson processes with rates λ_1 and λ_2, then $N(t) = N_1(t) + N_2(t)$ defines a Poisson process with rate $\lambda_1 + \lambda_2$.

Thinning. Each event of a Poisson process $\{N(t), t \geq 0\}$ with rate λ is classified as type I with probability p and as type II with probability $1 - p$, independently of all other events. The numbers of type I and type II events in $[0, t]$, $N_1(t)$ and $N_2(t)$, define independent Poisson processes with rates λp and $\lambda(1 - p)$.

Example. With $\lambda = 9$ claims per day and $p = \frac{1}{3}$, the type I and type II claims form independent Poisson processes with rates 3 and 6 per day.

Exercise 3.13: Two claim types: which type was this claim?

Exercise 3.13: [poi9, Solution] Ross: Chapter 5, Q59. There are two types of claims that are made to an insurance company. Let $N_i(t)$ denote the number of type i claims made by time t, and suppose that $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ are independent Poisson processes with rates $\lambda_1 = 10$ and $\lambda_2 = 1$. The amounts of successive type 1 claims are independent exponential random variables with mean \$1000 whereas the amounts from type 2 claims are independent exponential random variables with mean \$5000. A claim for \$4000 has just been received; what is the probability it is a type 1 claim.

Exercise 3.14: Portfolio takeover: liquid assets before and after

Exercise 3.14: [poi10, Solution] Insurer A has a combined home insurance and landlords insurance portfolio. The total number of claims for this portfolio is modelled using a Poisson process with expected claims 300 per year. The proportion of landlords insurance claims was one fifth of the overall claims. Insurer A sells its home insurance portfolio to insurer B. Insurer B specializes in home insurance and has no landlords insurance policies. The expected number of claims for the old portfolio of insurer B was 120 and can also be modelled using a Poisson process. The number of claims for insurer A is independent of the number of claims for insurer B.

1. Define the processes of the number of claims from insurer A for both the home insurance portfolio and landlords insurance portfolio before the takeover.
2. Define the processes of the number of claims of both insurer A and B before and after the takeover of the home insurance by insurer B.

Both insurers have to hold a liquid assets such that they are able to pay the claims arriving in a month with at least 99.9% certainty. Assume that each claim has a value of \$10,000.

1. Find the minimum amount of liquid assets that both insurers should hold before and after the takeover. (use Formualae and Tables pages 175-185 - approximate using normal distribution if needed and appropriate)
2. Comment on the takeover from the point of view of the liquid assets the insurers has to hold.

Common Mistakes

Not this	**This**
S_n taken as exponential	$T_n \sim \text{Exp}(\lambda)$ and $S_n \sim \text{Gamma}(n, \lambda)$
given $N(t) = n$, the count in a sub-interval taken as Poisson	$\text{Bin}(n, u/t)$ for a sub-interval of length u
mean $\lambda(s)\, t$ for $N(s+t) - N(s)$ in a non-homogeneous process	mean $m(s+t) - m(s)$
$N_1(t)$ and $N_2(t)$ from thinning taken as dependent	they are independent, although $N_1(t) + N_2(t) = N(t)$

Thanks for coming!